CSA5112 - Cryptography and Network Security

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1) Caesar cipher and Vigenere cipher :-

a(i) Caesar cipher :-

P = MEETATNOON

Key = 7.

Encryption formula :-

$$C = P + k \pmod{26}.$$

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M

13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

for M :- $C = P + k \pmod{26}$

$C = 12 + 7 \pmod{26}$

$C = 19 \mod 26 \Rightarrow 19 = T.$

for E :- $C = 4 + 7 \pmod{26}$

$C = 11 \pmod{26} \Rightarrow 11 = L.$

for E :- $C = 4 + 7 \pmod{26}$

$= 11 \pmod{26} \Rightarrow 11 = L.$

for T :- $C = 19 + 7 \pmod{26}$

$= 26 \mod 26 \Rightarrow 0 = A.$

for A :- $C = 0 + 7 \pmod{26} \Rightarrow 7 = H.$

for T :- $C = 19 + 7 \pmod{26} \Rightarrow 0 = A.$

for N :- $C = 13 + 7 \pmod{26} \Rightarrow 20 = U.$

for O :- $C = 14 + 7 \pmod{26} \Rightarrow 21 = V.$

for O:- $C = 14 + 7 \pmod{26} \Rightarrow 20 = V$.

for N:- $C = 13 + 7 \pmod{26} \Rightarrow 20 = U$.

plain text = MEETATNOON.

Ciphere text = TLLA HAUVVU.

(ii) Vigenere Cipher:-

plaintext: MEETATNOON

key = SUN

12	4	4	19	0	19	13	14	14	13
M	E	E	T	A	T	N	O	O	N
S	U	N	S	U	N	S	U	N	S
18	20	13	18	20	13	18	20	13	18.

for Encryption formula:-

$$C_i = k_i + P_i \pmod{26}.$$

for M:- $C = 18 + 12 \pmod{26} = 30 \bmod 26 \Rightarrow 4 = E$

for E:- $C = 20 + 4 \pmod{26} = 24 \bmod 26 \Rightarrow 24 = Y$.

for E:- $C = 13 + 4 \pmod{26} = 17 \bmod 26 \Rightarrow 17 = R$.

for T:- $C = 18 + 19 \pmod{26} = 37 \bmod 26 \Rightarrow 11 = L$.

for A:- $C = 20 + 0 \pmod{26} = 20 \bmod 26 \Rightarrow 20 = U$.

for T:- $C = 19 + 13 \pmod{26} = 32 \bmod 26 \Rightarrow 6 = G$.

for N:- $C = 18 + 13 \pmod{26} = 31 \bmod 26 \Rightarrow 5 = F$.

for O:- $C = 20 + 14 \pmod{26} = 34 \bmod 26 \Rightarrow 8 = I$.

for O:- $C = 13 + 14 \pmod{26} = 27 \bmod 26 \Rightarrow 1 = B$.

for N:- $C = 18 + 13 \pmod{26} = 31 \bmod 26 \Rightarrow 5 = F$.

plain text:- MEETATNOON

Cipher text:- EYRLUGFIBF.

Decryption formula:-

Cipher text:- EYRLUGFIBF.

key: SUN.

4	24	17	11	20	6	5	8	1	5
E	Y	R	L	U	G	F	I	B	F
S	U	N	S	U	N	S	U	N	S
18	20	13	18	20	13	18	20	13	18

$$P_i = C_i - K_i \pmod{26}$$

for E:- $P = 4 - 18 \pmod{26} \Rightarrow 12 = M$

for Y:- $P = 24 - 20 \pmod{26} \Rightarrow 4 = E$

for R:- $P = 17 - 13 \pmod{26} \Rightarrow 4 = E$

for L:- $P = 11 - 18 \pmod{26} \Rightarrow 19 = T.$

for U:- $P = 20 - 20 \pmod{26} \Rightarrow 0 = A.$

for G:- $P = 6 - 13 \pmod{26} \rightarrow 19 = T.$

for F:- $P = 5 - 18 \pmod{26} \Rightarrow 13 = N$

for I:- $P = 8 - 20 \pmod{26} \Rightarrow 14 = O$

for B:- $P = 1 - 13 \pmod{(26)} \Rightarrow 14 = O$

for f:- $P = 5 - 18 \pmod{26} \Rightarrow 13 = N.$

EYRLUGFIBF = MEETATNOON.

Decryption formula for Ceaser cipher:-

Cipher text:- TLLAHAUVVU.

key:- 7

$$P_i = C_i - K \pmod{26}.$$

for T :- P = 19 - 7 (mod 26) ⇒ 12 = M

for L :- P = 11 - 7 (mod 26) ⇒ 4 = E

for A :- P = 0 - 7 (mod 26) ⇒ 19 = T.

for H :- P = 7 - 7 (mod 26) ⇒ 0 = A

for U :- P = 20 - 7 (mod 26) ⇒ 13 = N.

for V :- P = 21 - 7 (mod 26) ⇒ 14 = O.

TLLA HA UVVU = MEETATNOON.

b) → In ceaser cipher, if there are any repeated characters (no. of accurances) in the plain text, it will be replaced by same cipher text characters.

→ In vigenere cipher, if there are any repeated characters occurs in the plain text, the cipher text will be unique from one another,

because if the key size is smaller than the plaintext it will be repeated upto the end of plaintext or else if the key size is larger then it take the character only with the plaintext size.

c) Vulnerabilities of analysis :-

Comparsion :-

Ceaser cipher	Vigenere cipher
1) It has 26 shifts	1) It has 26 unique key character for each plaintext.
2) Less secure compare with vigenere cipher.	2) More secure compared with ceaser cipher.

3) No. of occurances will be replaced by same characters. 4) It is substitution cipher.	3) Unique characters are replaced even if we have same character in plain text. 4) It is polyalphabetic cipher.

Conclusion:- By comparing both the ciphers, I recommend vigenere cipher to use because it is more secure to use as it is somewhat difficult to retrieve data from the user as compared with ceaser cipher.

2) Hill cipher:-

Given; Plain text = MATH

$$\text{key} = 2 \times 2 = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}.$$

Encryption formula:-

$$C = KP \pmod{26}.$$

From plain text:- MA | TH

$$\begin{bmatrix} M \\ A \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \end{bmatrix} \; ; \; \begin{bmatrix} T \\ H \end{bmatrix} = \begin{bmatrix} 19 \\ 7 \end{bmatrix}.$$

for $\begin{bmatrix} M \\ A \end{bmatrix}$; $C = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 12 \\ 0 \end{bmatrix} \pmod{26}$

$$= \begin{bmatrix} 36+0 \\ 60+0 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 36 \\ 60 \end{bmatrix} \pmod{26} \rightarrow \begin{bmatrix} 10 \\ 8 \end{bmatrix} = \begin{bmatrix} K \\ I \end{bmatrix}.$$

for $\begin{bmatrix} T \\ H \end{bmatrix}$; $C = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 19 \\ 7 \end{bmatrix} \bmod 26$

$$= \begin{bmatrix} 57 + 14 \\ 95 + 49 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 71 \\ 144 \end{bmatrix} \bmod 26 = \begin{bmatrix} 19 \\ 14 \end{bmatrix} = \begin{bmatrix} T \\ O \end{bmatrix}$$

(P) MATH = kITO. (C).

Decryption formula :-

Cipher text = kI|TO .

$$\text{key} = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$$ (a, b, c, d).

$$P = k^{-1} C \pmod{26}.$$

$kI/TO \rightarrow \begin{bmatrix} K \\ I \end{bmatrix} = \begin{bmatrix} 10 \\ 8 \end{bmatrix}$; $\begin{bmatrix} T \\ O \end{bmatrix} = \begin{bmatrix} 19 \\ 14 \end{bmatrix}$.

$$k^{-1} = \frac{1}{\det k} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

$\det k = 21 - 10 = 11$. $\Rightarrow 1/\det = 1/11$.

$k^{-1} \Rightarrow 11^{-1} \bmod 26 = 1$.

$11 \times 19 \bmod 26 = 1$.

$\boxed{x = 19}$ $\quad 11 \times x \bmod 26 = 1$.

$1/\det k = 19$.

$$k^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix}$$

(A)

$$k^{-1} = \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \bmod 26$$

$$k^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

for $\begin{bmatrix} K \\ I \end{bmatrix}$; $P = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 10 \\ 8 \end{bmatrix} \bmod 26$

$$= \begin{bmatrix} 30 + 112 \\ 90 + 40 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 142 \\ 130 \end{bmatrix} \bmod 26 \Rightarrow \begin{bmatrix} 12 \\ 0 \end{bmatrix} = \begin{bmatrix} M \\ A \end{bmatrix}.$$

for $\begin{bmatrix} T \\ O \end{bmatrix}$; $P = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 19 \\ 14 \end{bmatrix} \bmod 26$

$$= \begin{bmatrix} 57 + 196 \\ 171 + 70 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 253 \\ 241 \end{bmatrix} \bmod 26 \Rightarrow \begin{bmatrix} 19 \\ 7 \end{bmatrix} = \begin{bmatrix} T \\ H \end{bmatrix}.$$

$KITO = MATH.$

Major Weakness :-

Hill cipher is a polyalphabetic cipher, as compared to monoalphabetic ciphers, it can more securely encrypt the data, but with comparison with modern encryption techniques it is less secure, can be vulnerable to know the plain text (using) attacks.

b) Decryption of 'MATH' :-

Ciphertext :- MATH.

$$\text{key} = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}.$$

$$P = k^{-1} C \mod 26.$$

$$k^{-1} = 1/\det k \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Encryption formula:-

$$C = KP \mod 26$$

MA/TH ; $\begin{bmatrix} M \\ A \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \end{bmatrix}$; $\begin{bmatrix} T \\ H \end{bmatrix} = \begin{bmatrix} 19 \\ 7 \end{bmatrix}$.

for $\begin{bmatrix} T/H \end{bmatrix}$, $C = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}\begin{bmatrix} 19 \\ 7 \end{bmatrix} \mod 26$

$$= \begin{bmatrix} 38 + 21 \\ 76 + 35 \end{bmatrix} \mod 26.$$

$$= \begin{bmatrix} 59 \\ 111 \end{bmatrix} \mod 26 \Rightarrow \begin{bmatrix} 7 \\ 7 \end{bmatrix} = \begin{bmatrix} H \\ H \end{bmatrix}.$$

for $\begin{bmatrix} M/A \\ T/H \end{bmatrix}$; $C = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}\begin{bmatrix} 12 \\ 0 \end{bmatrix} \mod 26$

$$= 38 + \begin{bmatrix} 24 + 0 \\ 48 + 0 \end{bmatrix} \mod 26$$

$$= \begin{bmatrix} 24 \\ 48 \end{bmatrix} \mod 26 \Rightarrow \begin{bmatrix} 24 \\ 22 \end{bmatrix} = \begin{bmatrix} Y \\ W \end{bmatrix}.$$

MATH = YWHH.

Decryption formula:-

$$P = k^{-1} C \mod 26.$$

5

$$k^{-1} = 1/\det k \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$\det k = (10 - 12) = -2$.

Here, the det k value is negative (-2)..

So, it is not possible to calculate the k^{-1}..

3) AES - 256 GCM (vs) RSA key - exchange algorithm:-

Introduction:-

A hospital network of 200 clinics must send large MRI archieves (1GB each) to a central radiology server. For data encryption they have used AES - 256 algorithm and RSA for exchange of key securely..

Comparision:-

Feature	AES - 256 GCM	RSA
Type	It is Symmetric key encryption algorithm.	It is a Assymetric key exchange algorithm.
Speed of working	Very fast for large datasets / files.	Slow for large file.
Storage	Can be able to store /encrypt 1GB MRI archieves.	It is purely based on the key size
Size.	It is a 256 -bit secure key.	It is used to work the function of AES secret key.
Modern data analysis	Suitable	Less suitable.
key exchange	public key is shared securely	Private key will not be shared.

b) Hybrid encryption workflow :-

Hybrid encryption to means (the) here in this it combines both the AES - 256 GCM and RSA to (ach) archieve to clinical data.

→ Firstly it genereates a public key with a size of 256 - bit.

→ Over 19,900 keys are shared by using the formula:-

$$n(n-1)/2 = 200(199)/2 = 19,900.$$

, as per the 200 clinical repoits.

→ Encrypts the data using AES algorithm.

→ & Genereates the authentication tags and checks the integrity of the data.

Decryption :-

→ After encryption, it uses to RSA to decrypt the data.

→ Verify the authentication tags.

→ Here, the RSA - OAEP (OAEP) ~~optimi~~. is used for data encryption key wrapping.

→ The recieven has the shared public key to decrypt the data using the RS AES - 256 GCM algorithm.

Here, the algo modern algorithms are used so, that only the authenticated people can see and verify the clinical records.

4)

RSA demonstration :-

Given;

$p = 11$; $q = 19$

$e = 7$; $M = 5$.

(i) Calculate modulus n :-

$n = pq$

$n = 11 \times 19 = 209$.

(ii) $\phi(n)$:-

$\phi(n) = (p-1)(q-1)$

$= (10)(18) = 180$.

$\gcd(e, \phi(n)) = 1$

$7, 180 = 1$

Yes, 'e' is valid.

To find exponent d :-

$e \times d \equiv 1 \mod (\phi(n))$.

$7 \times d \equiv 1 \mod (180)$. $\Rightarrow 7 \times 103 \pmod{26}$.

$d = 103$

26×7
182
25×7
175.

Public key :- $(e, n) = (7, 209)$

Private key :- $(d, n) = (\ , 209)$

For encryption :- $(M = 5)$

$C = M^e \mod n$.

$C = 5^7 \mod 209$.

$C = 78,125 \mod 209$.

$\boxed{C = 168}$.

for Decryption :-

$$M = c^d \bmod(rs).$$
$$= 168^{103} \bmod 209.$$
$$\boxed{M = 5}.$$

→ Modern encryption technique RSA is used for sharing of the public key among user and reciever.

→ As it is a symmetic key, it uses private and public key for encryption and decryption.

RSA - OAEP :- Secure data Encryption.

RSA - PSS :- For secure key exchange.

5) Caesar cipher :-

Let us take an example of plain text :-

plain text :- HELLO. ; key = 3.

Encryption formula :- $C = k + p \pmod{26}$.

for H :- $C = 3 + 7 \pmod{26} = 10 \bmod 26 \Rightarrow 10 = K$

for E :- $C = 3 + 4 \pmod{26} = 7 \bmod 26 \Rightarrow 7 = H$

for L :- $C = 3 + 11 \pmod{26} = 14 \bmod 26 \Rightarrow 14 = O$

for L :- $C = 3 + 11 \pmod{26} = 14 \bmod 26 \Rightarrow 14 = O$

for O :- $C = 14 + 3 \pmod{26} = 17 \bmod 26 \Rightarrow 17 = R$

HELLO → KHOOR.

Decryption formula :- KHOOR = plaintext ; key = 3.

for K :- $P = 10 - 3 \bmod 26 = 7 \bmod 26 \Rightarrow 7 = H$

for H :- $P = 7 - 3 \bmod 26 = 4 \bmod 26 \Rightarrow 4 = E$

for O :- $P = 14 - 3 \bmod 26 = 11 \bmod 26 \Rightarrow 11 = L$

for O :- $P = 14 - 3 \bmod 26 = 11 \bmod 26 \Rightarrow 11 = L$

for R :- $P = 17 - 3 \bmod 26 = 14 \bmod 26 \Rightarrow 14 = O$.

KHOOR → HELLO.

Vigenere cipher :-

7	4	11	11	14
H	E	L	L	O
K	E	Y	K	E
10	4	24	10	4

Encryption formula :- $Ci = Ki + Pi \bmod 26$.

for H :- $C = 7 + 10 \bmod 26 = 17 \bmod 26 \Rightarrow 17 = R$

for E :- $C = 4 + 4 \bmod 26 = 8 \bmod 26 \Rightarrow 8 = I$

for L :- $C = 11 + 24 \bmod 26 = 9 \bmod 26 \Rightarrow 9 = J$

for L :- $C = 11 + 10 \bmod 26 = 21 \bmod 26 \Rightarrow 21 = V$.

for O :- $C = 14 + 4 \bmod 26 = 18 \bmod 26 \Rightarrow 18 = S$

HELLO → RIJVS.

Decryption formula :- $Pi = Ki - Ci \bmod 26$.

for R :- $P = 17 - 10 \bmod 26 = 7 \bmod 26 \Rightarrow 7 = H$

for I :- $P = 8 - 4 \bmod 26 = 4 \bmod 26 \Rightarrow 4 = E$

for J :- $P = 24 - 9 \bmod 26 = 15 \bmod 26 \Rightarrow 11 = L$

for V :- $P = 9 - 24 \bmod 26 = 11 \bmod 26 \Rightarrow 11 = L$

for S :- $P = 18 - 4 \bmod 26 = 14 \bmod 26 \Rightarrow 14 = O$.

H	E	L	L	O	– plain text
17	8	9	21	18	
R	I	J	V	S	– cipher text.
K	E	Y	K	E	– key.
10	4	24	10	4	

RIJVS → HELLO.

7 21

$\begin{array}{r} 703 \\ 2 \\ \hline 7\ 21 \end{array}$

Playfair cipher:-

Plaintext :- HELLO.

key :- KEY.

plaintext pairing :-

HE LL O → no same characters should be there

HE LX LO. → 3 pairs.

key :-

K	E	Y	A	B
C	D	F	G	H
I	J	M	N	O
P	Q	R	S	T
U	V	W	X	Z

Encryption :-

for HE :- Hth row and Eth column :- (Reg Rectangle rule).

H will be replaced by D.

Eth row and Hth column :-

B is replacement of E.

HE → DB.

for LX :- Lth row and xth column :-

N is replacement of L.

xth row and Lth column :-

V is replacement of V.

LX → NV.

for LO :- LO → JN.

I L M N O

HE LX LO → DB NV JN.

Decryption:-

ciphertext :- DB NV IN.

key :- KEY.

for DB :- Dth row and Bth column (Rectangle rule):-

H is replacement of D.

Bth row and Dth column

E is replacement of B.

DB → HE.

for NV :- Nth row and Vth column:-

L is replacement of N.

Vth row and Xth column:-

X is replacement of V.

NV → LX.

for IN:-



IN → LO.

DB NV IN → HE LX LO.

AES - 256 GCM :-

AES - 256 GCM is used modern encryption process.

Encryption process :-

1) Generates a secure 256-bit key.

2) Generates a secure public nonce for Encrypting the data.

3) Encrypts the customer data using AES - 256 GCM.

4) Generates authentication tag and integrity.

5) The generated encrypted data transmits through a network.

Decryption:-

→ The Encrypted message data is recieved.

→ For decryption it uses a secret key by the RSA-OAEP which uses for secure key exchange.

→ Decrypts the customer data.

→ Recieves to the messager inbox without any vulnerable attack known by plain text.

Conclusion:-

All the three ceaser, playfair and vigenereare ciphers are cryptographic classic encryption algorithms which are not much suitable for modern financial datas.

Whereas, the AES-256 GCM is a modern encryption algorithm, is used to store the financial data as it is more confidential, authenticate and integrity. RSA - OAEP is used for ~~key~~ se secure key exchange in this algorithm.